

1. Administrative & Study Identification

Official Study Title: An Open-Label, Randomized, Controlled Phase 3 Study of Sigvotatug Vedotin in Combination with Pembrolizumab Compared with Pembrolizumab Monotherapy as First-Line Treatment in Participants with PD-L1 High ($\geq 50\%$ of Tumor Cells Expressing PD-L1), Locally Advanced, Unresectable, or Metastatic Non-Small Cell Lung Cancer (Be6A LUNG-02). **Brief Study Title:** This is a Study to Learn About How the Combination of the Study Medicines Sigvotatug Vedotin Plus Pembrolizumab Works in People With Non-small Cell Lung Cancer With High Levels of PD-L1. **Short Title/Acronym:** Be6A LUNG-02 **Protocol Number:** C5751003 **NCT Number:** NCT06758401 **Posting ID:** 590872609063007348 **Trial Status:** Recruiting **Sponsor/Company Name:** Pfizer **Investigational Product:** Sigvotatug Vedotin + Pembrolizumab **Phase of Development:** Phase 3 **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and effectiveness of the study medicine (sigvotatug vedotin) used with a standard therapy (pembrolizumab) compared to pembrolizumab alone as a first-line treatment in patients with advanced non-small cell lung cancer (NSCLC). **Secondary Objectives:** To evaluate Progression-Free Survival (PFS) assessed by Investigator, Objective Response Rate (ORR) assessed by blinded independent central review (BICR), and Objective Response Rate (ORR) assessed by Investigator.

2.2 Background & Rationale To compare how the new combination treatment works compared to pembrolizumab alone in patients with non-small cell lung cancer (NSCLC) with high levels of PD-L1. PD-L1 is a protein that acts as a "brake" to keep the body's immune responses under control. This study targets people who have not yet been treated for advanced NSCLC (Stage 3 or 4) to determine if combining the antibody-drug conjugate sigvotatug vedotin with standard immunotherapy improves clinical outcomes.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Pembrolizumab monotherapy) **Blinding:** Open-label **Randomization:** Randomized (1:1 parallel group assignment)

3.2 Planned Enrollment & Duration Target NNS: 714 Participants
Number of Sites: Global multicenter **Estimated Study Start:** July 2025 **Estimated Primary Completion:** March 2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Have pathologically confirmed Stage IIIB or IIIC NSCLC and not be a candidate for surgical resection or definitive chemoradiation, or Stage IV NSCLC per the AJCC Staging Manual (Version 8.0) and the UICC Staging System (Eighth edition). 2. Participants with non-squamous histology must have documented negative test results for EGFR, ALK, and ROS1 AGAs and no known AGAs in NTRK, BRAF, RET, MET, or other AGAs with approved front-line therapies per local standard of care. Large cell neuroendocrine carcinoma is excluded. 3. Candidate for treatment with pembrolizumab monotherapy per local guidelines. 4. Tumor has PD-L1 expression in $\geq 50\%$ of tumor cells (TPS $\geq 50\%$) as determined by local testing. 5. Measurable disease based on RECIST v1.1 per investigator. 6. Resolution of acute effects of any prior therapy to either baseline severity or NCI CTCAE Grade 1 or less (except for AEs not constituting a safety risk in the investigator's judgment), unless otherwise excluded.

4.2 Exclusion Criteria 1. Life expectancy of < 3 months in the opinion of the investigator. 2. Any medical or psychiatric condition including recent (within the past year) or active suicidal ideation/behavior or laboratory abnormality that may increase the risk of study participation. 3. Any history of another malignancy within 3 years before the first dose of study intervention, or evidence of residual disease from a previously diagnosed malignancy. 4. Known or suspected hypersensitivity, intolerance, or contraindication to any excipient contained in the drug formulation of sigvotatug vedotin or pembrolizumab. 5. Respiratory conditions including: evidence of noninfectious or drug-induced ILD or pneumonitis; known DLCO (adjusted for hemoglobin) $< 50\%$ predicted; or Grade ≥ 3 pulmonary disease unrelated to underlying malignancy. 6. Known active CNS lesions. (Note: Definitively treated brain metastases or clinically inactive brain metastases of longest diameter < 0.5 cm are permitted). 7. Major surgery (requiring inpatient hospitalization ≥ 48 hours) within 21 days or minor surgery within 7 days prior to first dose. 8. Receipt of a live vaccine within 30 days prior to first dose. 9. Pre-existing peripheral neuropathy Grade ≥ 2 per NCI CTCAE v5.0. 10. Uncontrolled diabetes mellitus (HbA1c $\geq 8.0\%$ or HbA1c between 7.0% and 8.0% with associated symptoms like polyuria or polydipsia). 11. Prior immune-related AE that led to anti-PD-(L)1 treatment discontinuation, required a high-dose steroid taper (≥ 0.5 mg/kg prednisone) for > 2 weeks, or required systemic immunosuppressive therapy. 12. History of autoimmune disease that has required systemic treatment in the past 2 years. 13. Prior solid organ or bone marrow transplantation. 14. Currently

receiving a high-dose steroid (>10 mg prednisone per day) or condition requiring chronic immune suppressants. 15. Prior and concomitant therapy restrictions: No prior treatment with MMAE-derived drugs or IB6 targeting agents; no prior systemic therapy (including anti-PD-[L]1) for advanced NSCLC (neo-adjuvant/adjuvant allowed if recurrence occurred ≥ 9 months after last dose); no prior radiotherapy to the lung within 6 months; no chemotherapy/biologics completed less than 4 weeks prior; no prior therapy with an immune-oncology agent directed to a T-cell receptor. 16. History of or current ongoing infection, including active HIV, HBV, or HCV. 17. Severe uncontrolled cardiac or cerebrovascular condition within the previous 6 months.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: All participants receive pembrolizumab once every 6 weeks. Participants randomized to the combination arm also receive sigvotatug vedotin infusions 3 times per 6-week cycle (Day 1, Week 2, and Week 4, equivalent to once every 2 weeks). **Drug Details:** Pembrolizumab (Trade Name: Keytruda | ATC Code: L01FF02 | EMA URL: https://www.ema.europa.eu/en/documents/product-information/keytruda-epar-product-information_en.pdf). **Route of Administration:** Intravenous (IV) Infusion. **Duration of Treatment:** Pembrolizumab for up to about two years. Sigvotatug vedotin can continue until the NSCLC is no longer responding (disease progression) or until unacceptable toxicity occurs.

5.2 Concomitant & Prohibited Therapy Permitted: Definitively treated brain metastases or clinically inactive brain metastases < 0.5 cm. **Prohibited:** Prior therapy with an immune-oncology agent directed to a stimulatory or co-inhibitory T-cell receptor; chemotherapy/biologics completed less than 4 weeks prior to first dose; and prior radiotherapy to the lung within 6 months.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor/PD-L1 Assessment
Baseline	Day 0	Randomization, First Dose of Pembrolizumab ($\pm$ Sigvotatug Vedotin)
Interim	Q2W / Q6W	Regular clinic visits: IV Infusions, Toxicity monitoring, RECIST v1.1 Tumor Scans
End of Study	Up to 4 Years	Final Scans, Survival Follow-up, Exit Interview

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes 1. **Overall Survival (OS):** Defined as the duration from enrollment to death due to any cause. 2.

Progression-Free Survival (PFS) assessed by BICR: Defined as the time interval from the date of randomization to the date of first documented tumor progression determined by blinded independent central review (BICR) assessment as per RECIST 1.1 or death due to any cause, whichever comes first.

7.2 Secondary Outcomes 1. **Progression-Free Survival (PFS):** As assessed by Investigator. 2. **Objective Response Rate (ORR):** As assessed by BICR. 3. **Objective Response Rate (ORR):** As assessed by Investigator.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection via regular study clinic visits, physical examinations, and laboratory assessments monitoring for peripheral neuropathy, pneumonitis, and other toxicities. **Serious Adverse Events (SAEs):** Monitored and reported per standard regulatory timelines (typically 24 hours). **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee to assess ongoing safety and efficacy profiles.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy evaluation (OS/PFS); Safety Set for patients receiving at least one dose of the study drug. **Sample Size Justification:** Target enrollment of 714 participants powered to detect a statistically significant superiority in OS and PFS for the sigvotatug vedotin + pembrolizumab combination versus pembrolizumab alone. **Handling of Missing Data:** Censoring rules applied as per standard RECIST v1.1 progression guidelines and BICR charter.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to ensure protocol adherence and patient safety. **Audit Trail:** Validated electronic Data Capture (eDC) system with eCRF locking and query resolution workflows.

11. Study Identifiers & Governance

Primary ID: C5751003 **Posting ID:** 590872609063007348 **Registry Number:** NCT06758401 **Sponsor/Lead Organization:** Pfizer **Study Title:** Be6A LUNG-02 Clinical Trial **Trial Status:** Recruiting

12. Clinical Framework (Oncology Specific)

Primary Condition / Disease Areas: Non-Small Cell Lung Cancer; Carcinoma, Non-Small-Cell Lung; Carcinoma, Non-Small-Cell Lung (NSCLC). **UMLS / Semantic Framework:** * **Preferred UMLS Name:** Treatment of Non-Small Cell Lung Carcinoma * **Semantic Type:** Neoplastic Process * **Condition Entity:** Carcinoma (A malignant tumor arising from epithelial cells. Carcinomas that arise from glandular epithelium are called adenocarcinomas, those that arise from squamous epithelium are called squamous cell carcinomas...) * **MeSH Code:** D002277 * **HPO Code:** HP:0030731 * **SNOMED ID:** 1187425009 (Hierarchy links: 1187225007 | Malignant epithelial neoplasm, 118285006 | Epithelial neoplasm, 108369006 | Neoplasm, 1240414004 | Malignant neoplasm) **Diagnosis Threshold:** Stage IIIB, IIIC, or IV, PD-L1 Tumor Proportion Score $\geq 50\%$. **Medical Methodology:** Antibody-Drug Conjugate (ADC) + Anti-PD-1 Immunotherapy combination. **Optimization Target:** Overall Survival and prolonged Progression-Free Survival.

13. Study Procedures & Methodology

Management Pathway: First-Line standard of care compared against novel targeted combination. **Education Protocol (HCP):** RECIST v1.1 evaluation criteria training and AE management (e.g., ILD/pneumonitis). **Education Protocol (Patient):** Trial expectations, IV infusion schedule adherence, and reporting of immune-mediated adverse events. **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for unbiased tumor progression tracking.

14. Observation & Follow-Up Schedule

Total Duration: Up to approximately 4 years **Onsite Visit Cadence:** Day 1, Week 2, and Week 4 of every 6-week cycle **Radiological Monitoring Frequency:** Standard protocol intervals until disease progression **Checklist Review Interval:** Continuous throughout the treatment phase and post-progression survival follow-up.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	07-Apr-2026				
Clinical Operations Lead	[Name]	_____	07-Apr-2026		Lead		
Statistician	[Name]	_____	07-Apr-2026				